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St. Joseph's college f Arts & Science (Autonomous)

St. Joseph's College Road, Cuddalore–607001

B. Sc DEGREE EXAMINATION-APRIL 2021

EBC 510- PLANT BIOCHEMISTRY

Time: 3 hrs **Max Marks: 75**

SECTION–A (25X1=25)

Answer ALL the questions

I choose the best answer: (10X1=10)

1. Cellular organelles containing hydrolytic enzymes are called
   a) Peroisomes b) lysosomes c) ribosomes d) merosomes
2. Active uptake of minerals by roots mainly depends on the
   a) Availability of oxygen b) temperature c) light d) availability of co_2
3. The term auxin was coined by
   a) Skoog b) Haberlandt c) miescher d) F.W Went
4. All of the following hormones are growth inhibitors expect
   a) Absicic acid b) dormin c) ethylene d) IAA
5. Who was the first to study the influence of light during photosynthegi's?
   a) Van Niel b) Blackmann c) Warburg d) J. Ingen houz
6. The percentage of light energy fixed in photosynthesis is generclly around
   a) 0.1% b) 1% c) 10% d) 100%
7. The conversion of nitrogen to ammonic or nitrogenous compounds is called as
   a) Nitrogen assimilation b) Nitrogen fixation
   d) denitri fixation d) nitri fixation

8. Cyto kinins are

a) adenire derivatives b) guanine derivatives

c) cytidine derivatives d) thymine derivative

9. In nitrogen cycle nitrite is converted to nitrate by

a) Azotobacter b) Rhizobium c) Nitrobacter d) Clostridium

10. Which of the following N_2 fixer is involved in symbiotic association with legumes forming root nodules?

a) Rhizobium b)Azotobacter c) Rhodospirillum d) Clostridium

II. True or False (5X1=5)

11. Mitochodria is referred as power house of the cell
12. Active absorption involves use of ATP.
13. Auxin is a plant growth hormone.
14. Terpenoids are sometimes called isoprenoids.
15. Pectin is a plant secondary metabolite.

III. Answer in One or Two Sentences (10X1=10)

16. Define plant cell?
17. What is active absorption?
18. What is IAA?
19. What is Hill reaction?
20. Mention the role of cytokinin.
21. What are Carotenoids?
22. What are tannins?
23. Mention the importance of terpenoids.

24. Define photosynthesis?
25. Mention the importance of nitrogen fixation.

SECTION-B (4X5=20)

Answer any FOUR of the following:

26. Write a short note on ion exchange passive absorption.
27. Explain the structure, biosynthesis, mode of action and physiological effects of auxins.
28. Write a short note on structure and synthesis of chlorophyll.
29. Give an account of cyclic photophosphorylation.
30. How are plant secondary metabolites classified?
31. Give an account on glyoxylate cycle.

SECTION–C (3X10=30)

Answer any THREE of the following:

32. Elaborate on the mechanism of absorption of minerals in plants.
33. Describe the structure, biosynthesis, mode of action and physiological effects of gibberellins and IAA
34. Explain the structure and synthesis of phycobilins and Carotenoids.
35. Write an essay on nitrogen fixation.
36. Give a detailed account on photosynthesis.

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